When Should You Stop Looking? The 37% Rule, the Parking Problem, and What Model-Free Agents Rediscover

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Abstract

The "37% rule" states that when choosing sequentially from n options you should reject the first $1/e$ of them and then take the next record. We revisit this folklore result, place it inside a common Markov-decision-process (MDP) skeleton alongside the parking problem of MacQueen and Miller (1960), and ask an empirical question: do model-free reinforcement-learning agents, told nothing about the theory, rediscover the optimal stopping thresholds from reward alone? Exact backward induction gives computable ground truth. Tabular Q-learning recovers the secretary optimum to within 1.4% regret at $n = 100$ and matches the parking optimum exactly at high occupancy. In contrast, transplanting the 37% rule onto the parking problem is catastrophic — its regret reaches 1602.0% when free spots are common — and a per-observation search cost pulls the optimal cutoff from 0.368 down to 0.130. The 37% rule is thus a fragile special case, not a universal law of search; agents that learn from experience track the true, problem-dependent threshold instead.

1 Introduction

You are looking for a parking spot, driving toward your destination. Each spot you pass is either free or taken, and once you drive past a free one you cannot return. Stop too early and you walk a long way; hold out for the perfect spot and you may overshoot and find nothing. When should you take what you have?

The popular answer invokes the *secretary problem*: examine and reject the first 37% of the options, then take the next one that beats everything seen so far. The cutoff $1/e \approx 0.368$ is genuinely optimal — but only for the exact problem the secretary story describes: a fixed, known number of candidates, no recall, no cost to looking, and a payoff that rewards *only* selecting the single best. The parking problem violates several of these assumptions, and the 37% answer is simply wrong for it.

Question. Two questions organise this paper. (i) How fragile is the 37% rule — how far does the optimal stopping threshold move as we relax the secretary assumptions toward the parking problem and toward costly search? (ii) If an agent is given only the reward signal and no theory, does it *rediscover* the correct threshold from experience?

Approach. We cast the classic secretary problem, the parking problem, and two variants (a per-observation cost; cardinal "full information") inside one MDP skeleton: observe, then `stop` or `continue`. For the secretary problem the optimal policy has a closed form; for the parking problem we obtain exact ground truth by backward induction. Against these references we run tabular Q-learning and a one-parameter policy-gradient (REINFORCE) learner.

Findings. (i) The 37% rule emerges cleanly only in the classic rank-only, cost-free limit: at $n = 1000$ the optimum sits at $x^* = 0.368$ with success 0.368, but for $n = 10$ it is already 0.300. (ii) Transplanted onto parking, the 37% rule is disastrous: regret up to 1602.0% at occupancy $p = 0.90$. (iii) A search cost of 0.0050 per look pulls the optimal secretary cutoff down to 0.130. (iv) Q-learning recovers the secretary optimum (1.4% regret at $n = 100$) and the parking optimum (exact at $p = 0.90$) from reward alone; the one-parameter policy-gradient agent finds the right regime but with higher variance.

Contributions. A unified MDP treatment of the stopping family with exact-DP ground truth; a quantification of how badly the 37% rule fails off its home turf; and an empirical study of what model-free agents recover.

2 Related Work

The secretary problem. The best-choice problem and its $1/e$ law were popularised in the 1960s; Ferguson [3] traces its tangled history and Freeman [4] surveys the many variants. Gilbert and Mosteller [5] solved the *full-information* version, in which cardinal values (not just

ranks) are observed and the optimal win probability rises to ≈ 0.580.

The parking problem. MacQueen and Miller [6] introduced the "optimal persistence" or parking formulation, where the payoff depends on how far one stops from a goal rather than on picking a unique best. It is the canonical example in which the 37% intuition breaks down, because the optimal rule is a distance threshold that depends on how often spots are free.

Optimal stopping and RL. Optimal stopping is a classical topic in dynamic programming [2]; more recently it has been attacked with reinforcement learning and neural function approximation, e.g. for option pricing [1]. Our aim is narrower and pedagogical: using problems small enough to solve exactly, we ask what *tabular*, assumption-free agents recover, and use the closed forms as ground truth rather than as training signal.

3 Problems and Optimal Policies

All four problems share one skeleton: observe items in sequence; at each item choose `stop` (commit, ending the episode) or `continue`. They differ in what is observed and how the terminal reward is scored.

3.1 Classic secretary (rank-only)

n candidates arrive in uniform random order; only relative ranks are visible; the reward is 1 iff the candidate committed to is the global best. The optimal policy rejects the first r candidates and then takes the first *record* (best-so-far). Its success probability is

$$P(r,n) = \frac{r}{n} \sum_{i=r+1}^{n} \frac{1}{i-1}, \tag{1}$$

and as $n \to \infty$ with $x = r/n$ this tends to $P(x) = -x \ln x$, maximised at $x^\star = 1/e$ with $P^\star = 1/e$. We also score *arbitrary* (possibly non-threshold) policies exactly: a per-position take-mask S wins with probability $\sum_{i \in S} \frac{1}{n} \prod_{j<i,\, j \in S} \frac{j-1}{j}$, using the independence of record events (Eq. 1 is the threshold special case).

3.2 Parking problem

Driving toward a destination, we pass spots at known distances $c_1, \ldots, c_T$ (we use $[k, \ldots, 1, 0, 1, \ldots, k]$). Each spot is *free* independently with probability p. At a free spot we park (pay its distance, stop) or skip; running off the end pays a penalty F. The expected remaining cost obeys the backward recursion

$$V_t = p \, \min(c_t, V_{t+1}) + (1-p)\, V_{t+1}, \qquad V_T = F, \tag{2}$$

so the optimal action at a free spot is to park iff $c_t \le V_{t+1}$. The threshold V_{t+1} is *per-step*, not a single distance: near the end of the street one accepts almost anything rather than fail. Equation 2 is exact and serves as ground truth for the agents.

3.3 Variants

Costly search. Each inspected candidate costs c; the payoff is $P(\text{win}) - c\,\mathbb{E}[\#\text{inspected}]$. As c grows the optimal cutoff shrinks below $1/e$. **Full information.** Cardinal values are observed; the optimal win probability rises toward ≈ 0.580 [5], showing that the 37% ceiling is an artefact of discarding magnitude information.

3.4 Agents

Neither agent is given the closed forms. *Tabular Q-learning* operates on the discrete decision states (record position for secretary; street step for parking) with two actions, an ϵ-greedy behaviour policy, and linearly annealed exploration and learning rate; we score its greedy policy exactly via the formulas above. *REINFORCE* learns a single threshold parameter passed through a sigmoid, so the learned scalar *is* the cutoff — a natural fit for the stationary secretary rule, but deliberately mis-specified for the non-stationary parking optimum, which we use to illustrate that the policy class, not just the learner, matters.

4 Experimental Setup

Everything is synthetic and config-driven; no external data or pretrained models are used. The local configuration sweeps pool sizes $n \in \{10, 100, 1000\}$ (agents trained on $\{10, 50, 100\}$, where the tabular state space stays dense), parking occupancies $p \in \{0.3, 0.5, 0.7, 0.9\}$ on a street of half-length $k = 8$ with failure penalty $F = 20$, and observation costs $c \in \{0, \ldots, 0.0050\}$ at $n = 100$. Monte-Carlo estimates use 5×10^4 episodes per point; agents train for up to 8×10^4 episodes per cell over five seeds, and we report means with bootstrap 95% confidence intervals. The `gpu_full.yaml` configuration widens every sweep and raises the episode and seed counts; no code changes are required.

Reproduce with `make all`: `run_montecarlo` writes the closed-form and simulated curves, `run_agents` trains and scores the learners, and `plot_stopping` regenerates every figure together with the numeric macros used in this text.

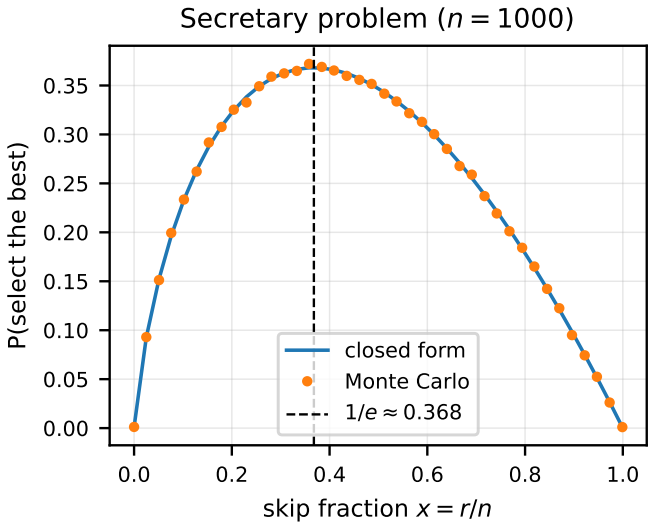


Figure 1: Secretary problem: Monte-Carlo success (points) matches the closed form $-x \ln x$ (line); the optimum is at $x = 1/e$.

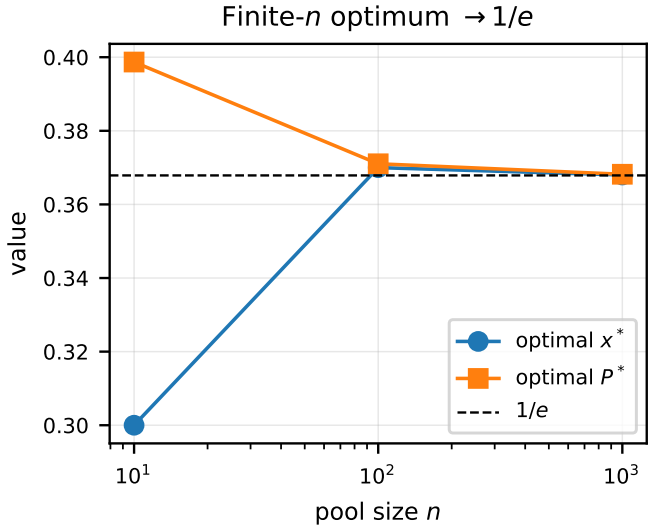


Figure 2: The optimal skip fraction x^* and success P^* approach $1/e$ only as $n \to \infty$.

5 Results

5.1 The 37% rule and its finite-n reality

Figure 1 shows the success probability against the skip fraction $x = r/n$ for $n = 1000$: the Monte-Carlo estimates lie on the closed-form curve $-x \ln x$, which peaks at $x = 1/e$. But the clean $1/e$ answer is asymptotic. Figure 2 tracks the optimum as n grows: at $n = 10$ the best cutoff is $x^* = 0.300$ with success 0.399 (higher than $1/e$, since fewer candidates make the best easier to catch), converging to $x^* = 0.368$, $P^* = 0.368$ only by $n = 1000$. The "37%" is a large-n limit, not a rule for small menus.

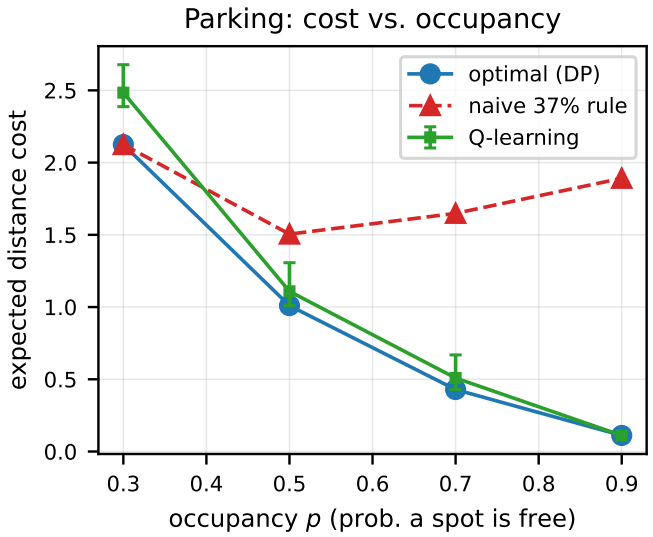


Figure 3: Parking: expected walking cost vs. occupancy p. The learned Q-agent tracks the DP optimum; the transplanted 37% rule does not.

5.2 The 37% rule fails on the parking problem

Figure 3 is the central result. The exact-DP optimum spends far less than the naive rule that blindly skips the first 37% of spots and then takes the first free one. The gap widens sharply with occupancy: at $p = 0.30$ the naive rule is essentially optimal (0.0% regret), but at $p = 0.90$ its regret explodes to 1602.0%, because when spots are usually free the right move is to hold out for the destination itself, not to grab the first free spot after an arbitrary cutoff.

5.3 Search cost pulls the cutoff below 37%

When looking is costly, patience is penalised. Figure 4 shows the optimal secretary cutoff falling monotonically from $1/e$ at $c = 0$ to 0.130 at $c = 0.0050$: with an inspection cost, one should commit earlier than the 37% folklore suggests.

5.4 Agents rediscover the thresholds

Given only rewards, tabular Q-learning recovers the secretary policy: Figure 5 shows its learned cutoff concentrating near $1/e$, and the greedy policy's exact success is within 1.4% of optimal at $n = 100$. On parking — the correct, per-step policy class — it matches the DP optimum exactly at $p = 0.90$ and stays within 16.9% in the sparse-occupancy regime, where the acceptance boundary is hardest to pin down from limited data. The one-parameter REINFORCE learner finds the right secretary regime but with markedly higher variance (17.6% regret), and by construction cannot represent the non-stationary parking optimum — evidence that recovering the rule requires a policy class rich enough to hold it.

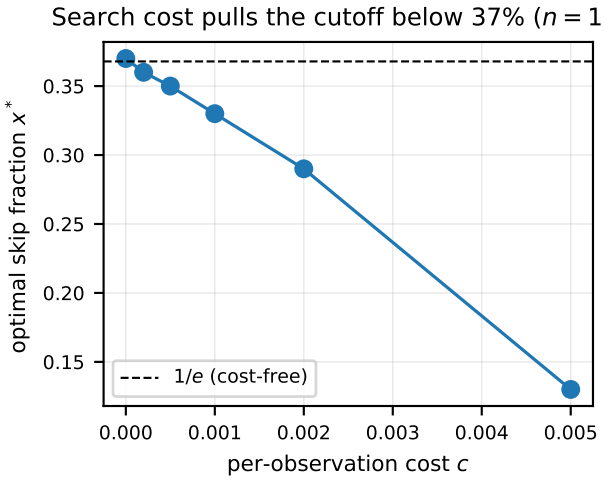


Figure 4: A per-observation cost pulls the optimal skip fraction below $1/e$.

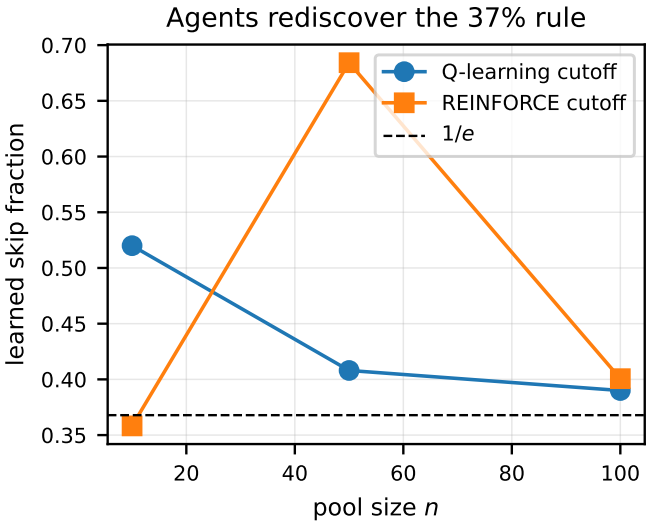


Figure 5: Learned skip fractions vs. pool size. Q-learning concentrates near $1/e$; the policy-gradient cutoff is noisier.

6 Discussion

When does the 37% rule actually apply? Our sweeps make the conditions concrete. The $1/e$ cutoff is close to optimal only when four things hold at once: the number of options is fixed and reasonably large; you observe ranks but care only about getting the single best; looking is free; and there is no recall. Loosen any one and the threshold moves — toward 0 under search cost, and toward the destination (hold out longer) as good options become common in the parking problem. Treating 37% as a universal heuristic for apartments, dates, or parking spots is therefore usually wrong; the quantity that transfers is the *method* (compute the continuation value and stop when the current option beats it), not the number.

What the agents tell us. The value-based agent recovers the optimal policy because its tabular representation can express the true, state-dependent threshold, and because exact scoring lets us read off regret rather than a noisy return. The policy-gradient agent's difficulty is instructive: a single learned cutoff is the right hypothesis class for the stationary secretary rule but cannot express the parking optimum at all. "Learning the rule from experience" thus decomposes into two requirements — enough data to estimate the continuation values, and a policy class rich enough to hold the answer.

Limitations. The problems are deliberately small so that exact ground truth exists; we do not claim these particular learners scale to high-dimensional stopping tasks. The parking model assumes independent occupancy and a known street; correlated availability and unknown length are natural next steps. The residual Q-learning regret in the sparse-occupancy regime reflects finite-sample estimation of the acceptance boundary, not a failure of the method, and shrinks under the larger-budget configuration.

7 Conclusion

The 37% rule is a beautiful but narrow result: exact for the classic secretary problem in the large-n, rank-only, cost-free limit, and badly wrong once any of those assumptions slips. Placing the secretary and parking problems in a common MDP with exact ground truth, we quantified the failure — naive regret up to 1602.0% on parking, and a cutoff dragged from 0.368 to 0.130 by a modest search cost — and showed that a tabular Q-learning agent, given only rewards, recovers the correct problem-specific threshold where a one-parameter policy-gradient learner cannot. The lesson for "when to stop looking" is not a magic percentage but a procedure: estimate the value of continuing, and stop the moment your current option beats it.

References

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